REGULARITY WITH LOCALLY NOETHERIAN FIBRES IS FPPF LOCAL ON THE TARGET AND SMOOTH LOCAL ON THE SOURCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5755-5761 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 5763-5799, the complete author proof (35 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Regular morphisms, fppf descent, and smooth base change for regularity; the reductions are retained.

Source reference. Original lemma descent-lemma-descending-property-flat: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5799–5799. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma descent-lemma-flat-fpqc-local-source: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5799–5799. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma descent-lemma-regular-local-smooth: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5799–5799. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-locally-Noetherian-fibres-fppf-local-source-and-target: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5578–5581. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma varieties-lemma-geometrically-regular-upstairs: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 5799–5799. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *The property $\mathcal{P}(f) =$"the fibres of f are locally Noetherian and f is regular" is local in the fppf topology on the target and local in the smooth topology on the source.*

Proof. We have $\mathcal{P}(f) = \mathcal{P}_1(f) \wedge \mathcal{P}_2(f) \wedge \mathcal{P}_3(f)$ where $\mathcal{P}_1(f) =$"the fibres of f are locally Noetherian", $\mathcal{P}_2(f) =$"f is flat", and $\mathcal{P}_3(f) =$"the fibres of f are geometrically regular". We have already seen that $\mathcal{P}_1$ and $\mathcal{P}_2$ are local in the fppf topology on the source and the target, see Lemma lemma-locally-Noetherian-fibres-fppf-local-source-and-target, and Descent, Lemmas descent-lemma-descending-property-flat and descent-lemma-flat-fpqc-local-source. Thus we have to deal with $\mathcal{P}_3$.

Let $f : X \to Y$ be a morphism of schemes. Let $\{\varphi_i : Y_i \to Y\}_{i \in I}$ be an fpqc covering of Y. Denote $f_i : X_i \to Y_i$ the base change of f by φ_i. Let $i \in I$ and let $y_i \in Y_i$ be a point. Set $y = \varphi_i(y_i)$. Note that

$$X_{i,y_i} = \mathrm{Spec}(\kappa(y_i)) \times_{\mathrm{Spec}(\kappa(y))} X_y.$$

Hence in this situation we have that X_y is geometrically regular if and only if X_{i,y_i} is geometrically regular, see Varieties, Lemma varieties-lemma-geometrically-regular-upstairs. This fact implies $\mathcal{P}_3$ is fpqc local on the target.

Let $\{X_i \to X\}$ be a smooth covering of X. Let $y \in Y$. In this case $\{X_{i,y} \to X_y\}$ is a smooth covering of the fibre. Hence the locality of $\mathcal{P}_3$ for the smooth topology on the source follows from Descent, Lemma descent-lemma-regular-local-smooth. Combining the above the lemma follows. □